

Why is my fundamental set of solutions/general solution of system of differential equations (SDE) different from the standard answer key?

Answer:

The short answer is: **Your solution is very likely still correct.**

A system of differential equations has an infinite number of possible fundamental sets of solutions. As long as your set is made of linearly independent solutions that solve the SDE, it is just as valid as the one in the answer key. They both describe the exact same set of all possible solutions (the general solution). Let's break down the specific reasons why your fundamental solutions might look different.

1. Choice of Eigenvectors (The Most Common Reason)

An eigenvector is not unique. If $\mathbf{v}$ is an eigenvector for an eigenvalue λ , then any non-zero scalar multiple $k\mathbf{v}$ is also a valid eigenvector for that same eigenvalue λ .

Different choices of scalar multiple of eigenvectors will lead to solutions that look different but are just scalar multiples of each other.

For larger $n \times n$ SDE, there may be repeated eigenvalues associated with more than one linearly independent eigenvectors. If $\mathbf{u}$ and $\mathbf{v}$ are eigenvectors for the same eigenvalue λ , then any linear combination $a\mathbf{u} + b\mathbf{v}$ is also a valid eigenvector for that same eigenvalue λ . Different choices of linear combinations of eigenvectors will lead to solutions that look different but are just linear combinations of each other.

2. Complex Eigenvalues

This case has two potential sources of difference.

- **Choice of Eigenvector:** Just like the real case, you can choose any scalar multiple of the complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$. Multiplying by a complex number $\alpha + \beta i$ will mix the real and imaginary parts $\mathbf{p}$ and $\mathbf{q}$, leading to a different-looking, but still valid, pair of real solutions $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$.
- **Choice of Eigenvalue:** The eigenvalues come in conjugate pair $\lambda_1 = a + ib$, $\lambda_2 = a - ib$ with corresponding eigenvectors $\mathbf{v}_1 = \mathbf{p} + i\mathbf{q}$ and $\mathbf{v}_2 = \mathbf{p} - i\mathbf{q}$. We have the choice of using a , b , $\mathbf{p}$, $\mathbf{q}$ in the formula of the real solutions $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, or using a , $-b$, $\mathbf{p}$, $-\mathbf{q}$ in the formula of $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$. The resulting $\mathbf{x}_{re}(t)$ will be the same, but the resulting $\mathbf{x}_{im}(t)$ will be negative of each other.

3. Repeated Eigenvalues (Generalized Eigenvectors)

When you have a repeated eigenvalue λ with only one linearly independent eigenvector $\mathbf{v}$, you need to find a generalized eigenvector $\mathbf{w}$ by solving $(\mathbf{A} - \lambda\mathbf{I})\mathbf{w} = \mathbf{v}$, the solution for $\mathbf{w}$ is not unique.

While the first solution $\mathbf{x}_1(t) = \mathbf{v}e^{\lambda t}$ is not affected, the second solution $\mathbf{x}_2(t) = (\mathbf{v}t + \mathbf{w})e^{\lambda t}$ depends on the choice of $\mathbf{w}$.

It is common to get different (but correct) answers for the fundamental and general solutions for the complex eigenvalue case and repeated eigenvalues case.

If you want to verify that your solution is correct, **plug your solution into the SDE:** This is the ultimate test. Take one of your fundamental solutions, say $\mathbf{x}_1(t)$. Calculate its derivative, $\mathbf{x}_1'(t)$. Then, separately, calculate the matrix product $\mathbf{A}\mathbf{x}_1(t)$. If $\mathbf{x}_1'(t) = \mathbf{A}\mathbf{x}_1(t)$, your solution is correct. Do this for both of your solutions.

Scenario 1 (Distinct Real Eigenvalues)

The system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ with matrix $\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 0 & 1 \end{pmatrix}$

1. We have two distinct eigenvalues: $\lambda_1 = -1$ and $\lambda_2 = -3$.

2. We have two corresponding eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (or any of their scalar multiples).
3. Our two solutions are $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t}$ and $\mathbf{x}_2(t) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-3t}$.
4. The general solution is $\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-3t} \quad (1)$$

In step 2 above, we may choose other scalar multiples of $\mathbf{v}_1$ and $\mathbf{v}_2$ as eigenvectors.

2. We use eigenvectors $\mathbf{u}_1 = \begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$ and $\mathbf{u}_2 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, where $\mathbf{u}_1 = \frac{1}{2} \mathbf{v}_1$ and $\mathbf{u}_2 = 2 \mathbf{v}_2$.
3. Our two solutions are $\mathbf{y}_1(t) = \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} e^{-t}$ and $\mathbf{y}_2(t) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-3t}$.
4. The general solution is $\mathbf{x}(t) = c_1 \mathbf{y}_1(t) + c_2 \mathbf{y}_2(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1/2 \\ 1 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 2 \\ 0 \end{pmatrix} e^{-3t} \quad (2)$$

We note that (2) can be written as $\mathbf{x}(t) = \frac{1}{2} c_1 \mathbf{x}_1(t) + 2 c_2 \mathbf{x}_2(t)$

Since (1) and (2) can both be expressed as linear combinations of $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$, they are equivalent to each other as general solution of the SDE.

Scenario 2 (Complex Eigenvalues)

The system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ with matrix $\mathbf{A} = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}$

1. We have two complex eigenvalues: $\lambda_1 = -1+2i$ and $\lambda_2 = -1-2i$.
2. We have two corresponding complex eigenvectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} - i \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
3. We use the eigen-pair λ_1 and $\mathbf{v}_1$ to get the real and imaginary parts: $a = -1$, $b = 2$ and $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Our two real solutions are

$$\begin{aligned} \circ \quad \mathbf{x}_{\text{re}}(t) &= e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos(2t) - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin(2t) \right] = e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix} \\ \circ \quad \mathbf{x}_{\text{im}}(t) &= e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin(2t) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos(2t) \right] = e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} \end{aligned}$$

4. The general solution is $\mathbf{x}(t) = c_1 \mathbf{x}_{\text{re}}(t) + c_2 \mathbf{x}_{\text{im}}(t)$:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} \quad (1)$$

In step 3 above, we may use the other eigen-pair:

3. We use the eigen-pair λ_2 and $\mathbf{v}_2$ to get the real and imaginary parts: $a = -1$, $b = -2$ and $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$.

Our two real solutions are

$$\begin{aligned} \circ \quad \mathbf{x}_{re}^*(t) &= e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos(-2t) - \begin{pmatrix} 0 \\ -1 \end{pmatrix} \sin(-2t) \right] = e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix} \\ \circ \quad \mathbf{x}_{im}^*(t) &= e^{-t} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin(-2t) + \begin{pmatrix} 0 \\ -1 \end{pmatrix} \cos(-2t) \right] = e^{-t} \begin{pmatrix} -\sin(2t) \\ -\cos(2t) \end{pmatrix} \end{aligned}$$

4. The general solution is $\mathbf{x}(t) = c_1 \mathbf{x}_{re}^*(t) + c_2 \mathbf{x}_{im}^*(t)$:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} \cos(2t) \\ -\sin(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -\sin(2t) \\ -\cos(2t) \end{pmatrix} \quad (2)$$

We note that (2) can be written as $\mathbf{x}(t) = c_1 \mathbf{x}_{re}(t) - c_2 \mathbf{x}_{im}(t)$

Since (1) and (2) can both be expressed as linear combinations of $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, they are equivalent to each other as general solution of the SDE.

It is also possible for us to use different eigenvectors by taking (complex) scalar multiples. For example, take $\mathbf{u}_1 = \begin{pmatrix} -i \\ 1 \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix} = -i \mathbf{v}_1$. This gives another eigenvector corresponding to $\lambda_1 = -1+2i$.

2. We use complex eigenvector $\mathbf{u}_1 = \begin{pmatrix} -i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.
3. The real and imaginary parts: $a = -1$, $b = 2$ and $\mathbf{p} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$.

Our two real solutions are

$$\begin{aligned} \circ \quad \mathbf{y}_{re}(t) &= e^{-t} \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos(2t) - \begin{pmatrix} -1 \\ 0 \end{pmatrix} \sin(2t) \right] = e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} \\ \circ \quad \mathbf{y}_{im}(t) &= e^{-t} \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin(2t) + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \cos(2t) \right] = e^{-t} \begin{pmatrix} -\cos(2t) \\ \sin(2t) \end{pmatrix} \end{aligned}$$

4. The general solution is $\mathbf{x}(t) = c_1 \mathbf{y}_{re}(t) + c_2 \mathbf{y}_{im}(t)$:

$$\mathbf{x}(t) = c_1 e^{-t} \begin{pmatrix} \sin(2t) \\ \cos(2t) \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -\cos(2t) \\ \sin(2t) \end{pmatrix} \quad (3)$$

We note that (3) can be written as $\mathbf{x}(t) = c_1 \mathbf{x}_{im}(t) - c_2 \mathbf{x}_{re}(t)$

Since (1) and (3) can both be expressed as linear combinations of $\mathbf{x}_{re}(t)$ and $\mathbf{x}_{im}(t)$, they are equivalent to each other as general solution of the SDE.

Scenario 3 (Repeated eigenvalues)

The system $\mathbf{x}' = \mathbf{A}\mathbf{x}$ with matrix $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$

1. We have a repeated eigenvalue: $\lambda = 2$.
2. We have a linearly independent eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (or any of its scalar multiples).
3. Our first solution is $\mathbf{x}_1(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t}$.
4. The generalized eigenvector is a solution of the non-homogeneous system $(\mathbf{A} - 2\mathbf{I})\mathbf{w} = \mathbf{v}$. This system

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

has infinitely many solutions given by the general solution $w_1 = 1 + s$ and $w_2 = s$ where s is a parameter. We can choose s to be any value to get a generalised eigenvector $\mathbf{w}$. For example, setting $s = 0$ gives $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We can also let $s = 1$ and get an alternative generalised eigenvector $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

5. If we choose $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, our second solution is $\mathbf{x}_2(t) = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) e^{2t}$ and the general solution is $\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} t+1 \\ t \end{pmatrix} e^{2t} \quad (1)$$

If we choose $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, our second solution is $\mathbf{x}_2^*(t) = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right) e^{2t}$ and the general solution is $\mathbf{x}(t) = c_1 \mathbf{x}_1(t) + c_2 \mathbf{x}_2^*(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} t+2 \\ t+1 \end{pmatrix} e^{2t} \quad (2)$$

The difference between (1) and (2) lies in the second term.

$$(2) = (1) + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} = (1) + c_2 \mathbf{x}_1(t)$$

We note that (2) can be written as $\mathbf{x}(t) = (c_1 + c_2) \mathbf{x}_1(t) + c_2 \mathbf{x}_2(t)$.

Since (1) and (2) can both be expressed as linear combinations of $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$, they are equivalent to each other as general solution of the SDE.

Let's go back to step 2 and choose a different eigenvector.

2. We have a linearly independent eigenvector $\mathbf{v} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ (or any of its scalar multiples).
3. Our first solution is $\mathbf{y}_1(t) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{2t}$.
4. The generalized eigenvector is a solution of the non-homogeneous system $(\mathbf{A} - 2\mathbf{I})\mathbf{w} = \mathbf{v}$. This system

$$\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

has infinitely many solutions given by the general solution $w_1 = 2 + s$ and $w_2 = s$ where s is a parameter. We can choose s to be any value to get a generalised eigenvector $\mathbf{w}$. For example, setting $s = 0$ gives $\mathbf{w} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$. We can also let $s = 1$ and get an alternative generalised eigenvector $\mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

5. If we choose $\mathbf{w} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$, our second solution is $\mathbf{y}_2(t) = \left(\begin{pmatrix} 2 \\ 2 \end{pmatrix} t + \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right) e^{2t}$ and the general solution is $\mathbf{x}(t) = c_1 \mathbf{y}_1(t) + c_2 \mathbf{y}_2(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 2t+2 \\ 2t \end{pmatrix} e^{2t} \quad (3)$$

If we choose $\mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, our second solution is $\mathbf{y}_2^*(t) = \left(\begin{pmatrix} 2 \\ 2 \end{pmatrix} t + \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right) e^{2t}$ and the general solution is $\mathbf{x}(t) = c_1 \mathbf{y}_1(t) + c_2 \mathbf{y}_2^*(t)$:

$$\mathbf{x}(t) = c_1 \begin{pmatrix} 2 \\ 2 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 2t+3 \\ 2t+1 \end{pmatrix} e^{2t} \quad (4)$$

We note that the difference between (3) and (4) lies in the second term.

$$(4) = (3) + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} = (1) + \frac{1}{2} c_2 \mathbf{y}_1(t)$$

We note that (4) can be written as $\mathbf{x}(t) = (c_1 + \frac{1}{2} c_2) \mathbf{y}_1(t) + c_2 \mathbf{y}_2(t)$.

Since (3) and (4) can both be expressed as linear combinations of $\mathbf{y}_1(t)$ and $\mathbf{y}_2(t)$, they are equivalent to each other as general solution of the SDE.

Furthermore, both terms in (3) are scalar multiple 2 of the corresponding terms in (1). So

$$(3) = 2c_1 \mathbf{x}_1(t) + 2c_2 \mathbf{x}_2(t)$$

$$(4) = (2c_1 + c_2) \mathbf{x}_1(t) + 2c_2 \mathbf{x}_2(t)$$

Since (3) and (4) can both be expressed as linear combinations of $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$, they are equivalent to (1) and (2) as general solution of the SDE.